

Part A

Problem 1: Phasor Representation

(a) $v_1(t) = 20 \sin(\omega t + \frac{\pi}{5}) \text{ V}$

Phasor: $\bar{V}_1 = 20 \angle \frac{\pi}{5} \text{ V}$

(b) $v_2(t) = -50 \sin(\omega t) \text{ V}$

Convert to sine:

$$-50 \sin(\omega t) = 50 \sin(\omega t + \pi)$$

Phasor: $\bar{V}_2 = 50 \angle \pi \text{ V}$

(c) $v_3(t) = 200 \cos(\omega t) \text{ V}$

Convert to sine:

$$200 \cos(\omega t) = 200 \sin(\omega t + \frac{\pi}{2})$$

Phasor: $\bar{V}_3 = 200 \angle \frac{\pi}{2} \text{ V}$

(d) $i_1(t) = 100 \sin(10^4 t - \frac{\pi}{3}) \text{ mA}$

Phasor: $\bar{I}_1 = 100 \angle -\frac{\pi}{3} \text{ mA}$

(e) $i_2(t) = -5 \cos(\omega t - \frac{\pi}{4}) \text{ A}$

Convert to sine:

$$-5 \cos(\omega t - \frac{\pi}{4}) = 5 \sin(\omega t - \frac{\pi}{4} + \frac{3\pi}{2})$$

Phasor: $5 \angle -\frac{3\pi}{4} \text{ A}$

(f) $i_3(t) = 50 \sin(100t + \frac{\pi}{4}) \text{ mA}$

Phasor: $\bar{I}_3 = 50 \angle \frac{\pi}{4} \text{ mA}$

Problem 2: Time-Domain Conversion

(a) $\bar{V}_1 = 100 \angle \frac{\pi}{4} \text{ mV}$

Time-domain: $v_1(t) = 100 \sin(\omega t + \frac{\pi}{4}) \text{ mV}$

(b) $\bar{V}_2 = 200 \angle 0 \text{ V}$

Time-domain: $v_2(t) = 200 \sin(\omega t) \text{ V}$

(c) $\bar{V}_3 = 10 \angle -\frac{\pi}{4} \text{ V}$

Time-domain: $v_3(t) = 10 \sin(\omega t - \frac{\pi}{4}) \text{ V}$

(d) $\bar{I}_1 = 100 \angle -\frac{\pi}{6} \text{ mA}$

Time-domain: $i_1(t) = 100 \sin(\omega t - \frac{\pi}{6}) \text{ mA}$

Problem 3: Phasor Addition

(a) $1.5 \cos(\omega t + 10^\circ) + 3.2 \cos(\omega t + 25^\circ)$

Convert to phasors:

$$1.5 \angle 10^\circ, \quad 3.2 \angle 25^\circ$$

Rectangular form:

$$(1.477 + 2.900) + j(0.260 + 1.352) = 4.377 + j1.612$$

Magnitude:

$$\sqrt{4.377^2 + 1.612^2} \approx 4.67$$

Phase:

$$\tan^{-1} \left(\frac{1.612}{4.377} \right) \approx 20.2^\circ$$

Final Answer:

$$\boxed{4.67 \cos(\omega t + 20.2^\circ)}$$

(b) $50 \sin(\omega t - 60^\circ) + 24 \sin(\omega t + 15^\circ)$

Convert to phasors:

$$50 \angle -60^\circ, \quad 24 \angle 15^\circ$$

Rectangular form:

$$(25 + 23.18) + j(-43.30 + 6.21) = 48.18 - j37.09$$

Magnitude:

$$\sqrt{48.18^2 + (-37.09)^2} \approx 60.8$$

Phase:

$$\tan^{-1} \left(\frac{-37.09}{48.18} \right) \approx -37.6^\circ$$

Final Answer:

$$\boxed{60.8 \sin(\omega t - 37.6^\circ)}$$

Problem 4: Current Amplitude and PhaseGiven $I_1(t) = 0.82 \cos(\omega t - 30^\circ)$ and $I_2(t) = 0.5 \cos(\omega t - 36^\circ)$:

$$\text{Sum: } 1.115 - j0.704$$

Magnitude:

$$\sqrt{1.115^2 + (-0.704)^2} \approx 1.32 \text{ A}$$

Phase:

$$\tan^{-1} \left(\frac{-0.704}{1.115} \right) \approx -32.3^\circ$$

Final Answer:

$$\boxed{1.32 \text{ A at } -32.3^\circ}$$

Part B

Problem 1: Series Impedances

(a) $R = 100 \Omega, L = 2 \text{ mH}$

$$\begin{aligned} Z_{\text{eq}} &= R + j\omega L = 100 + j(10^4 \times 2 \times 10^{-3}) \\ &= \boxed{100 + j2 \times 10^{-3} \omega \Omega} \end{aligned}$$

(b) $R = 100 \Omega, L = 0.1 \text{ H}, C = 1000 \mu\text{F}$

$$\begin{aligned} Z_{\text{eq}} &= R + j\omega L - \frac{j}{\omega C} \\ &= 100 + j0.1\omega - j\frac{1}{10^4 \times 10^{-3}} \\ &= \boxed{100 + j\left(0.1\omega - \frac{10^3}{\omega}\right) \Omega} \end{aligned}$$

(c) $R = 100 \Omega, C = 100 \mu\text{F}$

$$\begin{aligned} Z_{\text{eq}} &= R - \frac{j}{\omega C} = 100 - j\frac{1}{\omega \times 100 \times 10^{-6}} \\ &= \boxed{100 - j\frac{10^4}{\omega} \Omega} \end{aligned}$$

(d) $L = 2 \text{ mH}, C = 1000 \mu\text{F}$

$$\begin{aligned} Z_{\text{eq}} &= j\omega L - \frac{j}{\omega C} = j\left(\frac{\omega}{500} - \frac{1000}{\omega}\right) \\ &= \boxed{+j\left(\frac{\omega}{500} - \frac{1000}{\omega}\right) \Omega} \end{aligned}$$

Problem 2: Parallel Impedances ($\omega = 100 \text{ rad/s}$)

(a) $R = 100 \Omega, L = 1 \text{ H}$

$$\begin{aligned} Z_L &= j\omega L = j100 \\ Z_{\parallel} &= \frac{R \cdot Z_L}{R + Z_L} = \frac{100 \cdot j100}{100 + j100} \\ &= \frac{100^2 \angle 90^\circ}{100\sqrt{2} \angle 45^\circ} = \boxed{70.7 \angle \frac{\pi}{4} \Omega} \end{aligned}$$

(b) $R = 100 \Omega, L = 1 \text{ H}, C = 100 \mu\text{F}$

$$\begin{aligned} Z_C &= -j\frac{1}{\omega C} = -j100 \\ Y_{\text{total}} &= \frac{1}{R} + \frac{1}{Z_L} + \frac{1}{Z_C} = 0.01 - j0.01 + j0.01 \\ &= 0.01 \implies Z_{\parallel} = \boxed{100 \Omega} \end{aligned}$$

(c) $R = 100 \Omega, C = 100 \mu\text{F}$

$$\begin{aligned}
 Z_C &= -j100 \\
 Z_{\parallel} &= \frac{R \cdot Z_C}{R + Z_C} = \frac{100(-j100)}{100 - j100} \\
 &= \frac{100^2 \angle -90^\circ}{100\sqrt{2} \angle -45^\circ} = \boxed{70.7 \angle -\frac{\pi}{4} \Omega}
 \end{aligned}$$

(d) $L = 1 \text{ H}, C = 100 \mu\text{F}$

$$\begin{aligned}
 Z_L &= j100, \quad Z_C = -j100 \\
 Z_{\parallel} &= \frac{Z_L \cdot Z_C}{Z_L + Z_C} = \infty \implies \boxed{\infty}
 \end{aligned}$$

in parallel the **impedance** of L and C is product over sum**Problem 3: Series RLC** ($\omega = 20,000 \text{ rad/s}$)

$$\begin{aligned}
 Z &= R + j\omega L - \frac{j}{\omega C} \\
 &= 15 + j(20000 \times 10^{-3}) - j\frac{1}{20000 \times 10^{-6}} \\
 &= 15 + j20 - j50 = \boxed{15 - j30 \Omega = 33.5 \angle -63.4^\circ \Omega}
 \end{aligned}$$

Problem 4: Parallel RLC ($\omega = 20,000 \text{ rad/s}$)

$$\begin{aligned}
 Y &= \frac{1}{R} + \frac{1}{j\omega L} + j\omega C \\
 &= \frac{1}{15} - j0.05 + j0.02 = 0.0667 - j0.03 \\
 Z &= \frac{1}{Y} \approx \boxed{13.7 \angle 23^\circ \Omega}
 \end{aligned}$$

Problem 5: Equivalent Impedance Z_{AB} (a) **Series** 6Ω and $j2 \Omega$:

$$Z_{AB} = \boxed{6 + j2 \Omega = 6.32 \angle 0.32 \text{ rad} \Omega}$$

(b) **Parallel** 10Ω and $-j10 \Omega$:

$$Z_{AB} = \frac{10 \cdot (-j10)}{10 - j10} = \boxed{10\sqrt{2} \angle -\frac{\pi}{4} \Omega}$$

(c) $\omega = 100 \text{ rad/s}$:

$$Z_{AB} = 10 + j(100 \times 0.1) = \boxed{10 + j10 \Omega = 10\sqrt{2} \angle \frac{\pi}{4} \Omega}$$

(d) $f = 50 \text{ Hz}$:

$$\omega = 2\pi \times 50 \approx 314.16 \text{ rad/s} \quad Z_{AB} = 30 - j(314.16 \times 0.1) \approx \boxed{30 - j0.4 \Omega}$$

Problem 6: Network Impedance ($\omega = 3000 \text{ rad/s}$)

$$Z_1 = 3300 + j5700 \Omega, \quad Z_2 = 22000 - j49019.6 \Omega$$

$$Z_{\text{eq}} = \frac{Z_1 \cdot Z_2}{Z_1 + Z_2} \approx \boxed{7.05 \angle 53.8^\circ \text{ k}\Omega}$$

Part C**Problem 1: Voltage Across Components**

$$Z_{\text{total}} = 12 + j12 + 6 = 18 + j12 \Omega$$

$$I = \frac{25}{21.63} \angle -33.7^\circ \approx 1.155 \angle -33.7^\circ \text{ A}$$

$$V_{12} = 12 \cdot I = \boxed{13.96 \angle -33.7^\circ \text{ V}}$$

$$V_{6H} = j12 \cdot I = \boxed{13.96 \angle 56.3^\circ \text{ V}}$$

$$V_6 = 6 \cdot I = \boxed{6.95 \angle -33.7^\circ \text{ V}}$$

Problem 2: Finding $i(t)$, $v_1(t)$, and $v_2(t)$

Given $v_S(t) = 200 \sin(2\pi 50t)$:

$$\omega = 2\pi \times 50 = 100\pi \text{ rad/s}$$

$$Z_R = 10 \Omega$$

$$Z_L = j\omega L = j(100\pi)(0.1) = j31.416 \Omega \approx j31.416 \Omega$$

$$Z_{\text{total}} = Z_R + Z_L = 10 + j31.416 \Omega$$

$$|Z_{\text{total}}| = \sqrt{10^2 + 31.416^2} \approx 32.97 \Omega$$

$$\theta_Z = \tan^{-1} \left(\frac{31.416}{10} \right) \approx 1.26 \text{ rad (lagging)}$$

$$I_{\text{phasor}} = \frac{200 \angle 0^\circ}{32.97 \angle 1.26} \approx 6.06 \angle -1.26 \text{ A}$$

$$i(t) = 6.06 \sin(100\pi t - 1.26) \text{ A} \approx \boxed{6.1 \sin(\omega t - 1.26) \text{ A}}$$

$$v_1(t) = I_{\text{peak}} \times R \sin(\omega t - 1.26) = 6.06 \times 10 \sin(100\pi t - 1.26)$$

$$\approx \boxed{61 \sin(\omega t - 1.26) \text{ V}}$$

$$v_2(t) = I_{\text{peak}} \times \omega L \sin \left(\omega t - 1.26 + \frac{\pi}{2} \right)$$

$$= 6.06 \times 31.416 \sin(100\pi t + 0.31)$$

$$\approx \boxed{191 \sin(\omega t + 0.31) \text{ V}}$$

Problem 3: RMS Values of i , i_1 , and i_2

Given $V_S = 71 \text{ V (RMS)}$, $f = 50 \text{ Hz}$:

$$\begin{aligned}
 I_1 &= \frac{V_S}{R} = \frac{71}{10} = \boxed{7.1 \text{ A}} \\
 I_2 &= V_S \times (2\pi f C) \\
 &= 71 \times (2\pi \times 50 \times 100 \times 10^{-6}) \\
 &\approx \boxed{2.23 \text{ A}} \\
 I &= \sqrt{I_1^2 + I_2^2} \\
 &= \sqrt{(7.1)^2 + (2.23)^2} \\
 &\approx \boxed{7.44 \text{ A}}
 \end{aligned}$$

Problem 4: RMS Source Current

Given $V_S = 50 \text{ V (RMS)}$, $\omega = 300 \text{ rad/s}$:

$$\begin{aligned}
 Z &= (10 + j(0.1 \times 300)) \parallel \left(\frac{1}{j(300 \times 100 \times 10^{-6})} \right) \\
 &= (10 + j30) \parallel \left(\frac{1}{j0.03} \right) \\
 &= (10 + j30) \parallel (-j33.33) \\
 &= \frac{(10 + j30)(-j33.33)}{10 + j30 - j33.33} \\
 &= \frac{(10 + j30)(-j33.33)}{10 - j3.33} \\
 &= 100 \Omega \\
 I &= \frac{V_S}{Z} = \frac{50}{100} = \boxed{0.5 \text{ A}}
 \end{aligned}$$

Problem 5: Time-Domain Expressions for I_a , I_b , I_s

Given $V_S(t) = 340 \cos(377t)$, $Z_1 = 5.9 \angle 7^\circ \Omega$, $Z_2 = 2.3 \angle 0^\circ \Omega$, $Z_3 = 17 \angle 11^\circ \Omega$:

$$\begin{aligned}
 I_a &= \frac{V_s(t)}{Z_1 + Z_2} \\
 &= \frac{340 \cos(377t)}{5.9 \angle 7^\circ + 2.3 \angle 0^\circ} \\
 &= \frac{340 \angle 0^\circ}{5.92 \angle 7^\circ + 2.3 \angle 0^\circ} \\
 &\approx 41.5 \angle -5.05^\circ \\
 &= \boxed{41.5 \cos(377t - 5^\circ) \text{ A}}
 \end{aligned}$$

$$\begin{aligned}
 I_b &= \frac{V_s(t)}{Z_3} \\
 &= \frac{340 \cos(377t)}{17 \angle 11^\circ} \\
 &= \frac{340 \angle 0^\circ}{17 \angle 11^\circ} \\
 &= 20 \angle 79^\circ \\
 &= \boxed{20 \cos(377t - 11^\circ) \text{ A}}
 \end{aligned}$$

$$\begin{aligned}
 I_s &= I_a + I_b \\
 &= \frac{340 \angle 0^\circ}{5.92 \angle 7^\circ + 2.3 \angle 0^\circ} + 20 \angle 79^\circ \\
 &\approx 61.5 \angle 83^\circ \\
 &= \boxed{61.5 \cos(377t - 7^\circ) \text{ A}}
 \end{aligned}$$

Problem 6: Current Amplitude and Phase

(a) Given $Z = 8 + j6 \Omega$, $V_{\text{peak}} = 5 \text{ V}$:

$$\begin{aligned}
 |Z| &= \sqrt{8^2 + 6^2} = \sqrt{100} = 10 \Omega \\
 I_{\text{peak}} &= \frac{V_{\text{peak}}}{|Z|} = \frac{5}{10} = \boxed{0.5 \text{ A}}
 \end{aligned}$$

(b) Phase angle relative to voltage:

$$\begin{aligned}
 \theta_Z &= \tan^{-1} \left(\frac{6}{8} \right) \approx 36.9^\circ \\
 \phi_{\text{current}} &= -\theta_Z = \boxed{-36.9^\circ}
 \end{aligned}$$

Problem 7: Impedance and Component Values

(a) Given $V_{\text{RMS}} = 230 \text{ V}$, $I_{\text{RMS}} = 5 \text{ A}$:

$$Z = \frac{V_{\text{RMS}}}{I_{\text{RMS}}} = \frac{230 \angle 0}{5 \angle -\frac{\pi}{6}} = \boxed{46 \angle \frac{\pi}{6} \Omega} \quad (\text{lagging})$$

(b) Series R - L model:

$$\begin{aligned}
 R &= |Z| \cos(\theta) = 46 \cos(30^\circ) \approx \boxed{39.8 \Omega} \\
 X_L &= |Z| \sin(30^\circ) = 23 \Omega \\
 L &= \frac{X_L}{2\pi f} = \frac{23}{100\pi} \approx \boxed{73 \text{ mH}}
 \end{aligned}$$

Problem 8: Minimum Impedance Frequency

Given $R = 100 \Omega$, $L = 0.1 \text{ mH}$, $C = 1000 \mu\text{F}$:

$$\begin{aligned}
 f_0 &= \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1 \times 10^{-7}}} \approx \boxed{503 \text{ Hz}} \\
 Z_{\text{min}} &= R = \boxed{100 \Omega}
 \end{aligned}$$